DATA TYPES IN PYTHON

DATATYPES IN PYTHON

- It plays an important role in the definition of variables and function in statically oriented programming languages
- In python **everything is an object**. Data types are actually built-in classes.
- It specifies the type of data that a variable can store data such as int, float, char, string, etc
- Datatype is mainly used to store and manipulate the different types of data via variable
- Python has following built-in data types. They are
    1. String Types / Text Types
    2. Numeric Types
    3. Boolean Types
    4. None Types
    5. Sequence Types
    6. Mapping Type
    7. Set Types
    8. Binary Types

type(object)

- It is a built-in function in python which is used to return or display the type of an object / variable or create a new type object
- This function will take one argument which is the datatype of object in python
- Return type: new type

Example

```
type(data)
```

Data Types

S.N	Category	Data Types / Class Name
1.	String / Text	str
2.	Numeric Types	int, float, complex
3.	Boolean Types	bool
4.	None Type	None
5.	Binary Types	bytes, bytearray, memoryview
6.	Sequence Types	list, tuple, range
7.	Mapping Type	dict
8.	Set Types	set, frozenset

1. String Types (Text Types)

- This type is used to store the textual information
- Here, built-in type str is used to store both character as well as string
- In python, there is no specific data type for character.
- The string type in python is called as **str**.
- Here message can be embedded into python string using **single quote** or **double quotes** or **triple single quotes**.

Built-in Data Type

str

Example

```
d1=”hello World”
        (OR)
d1=’hello world’
        (OR)
d1=’’’hello world’’’
```

I. STRING TYPE EXAMPLE

1. SOURCE CODE

```
print("--------------------------------------")
print("\t String Type Example")
print("--------------------------------------")
# define the bytes variables with initial values
m1='Hello..Hi'
m2="Good Morning"
m3='''Have a Nice Day'''
# display the values
print(m1)
print(m2)
print(m3)
# display the data types (class types)
print("m1 Type: ",type(m1))
print("m2 Type: ",type(m2))
print("m3 Type: ",type(m3))
```

2. OUTPUT

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\IDE Projects\VSC Projects\Python Lab\ptmech> python st.py
---------------------------------------
         String Type Example
---------------------------------------
Hello..Hi
Good Morning
Have a Nice Day
m1 Type:  <class 'str'>
m2 Type:  <class 'str'>
m3 Type:  <class 'str'>
PS E:\IDE Projects\VSC Projects\Python Lab\ptmech>
```

2. Numeric Types

- This type of the data is generally used to store the numbers which can be integer, floating point numbers, complex numbers.
- It is classified as three types. They are
  - Integer
  - Float
  - Complex

int (Integer)

- This built-in type is used to store signed integers of unlimited length

float

- This built-in type is used to hold floating point numbers upto 15 decimal places

complex

- This datatype represents the complex number which contains the real and imaginary parts

Built-in Data Type

int, **float**, **complex**

Example

```
a=45          # storing integer value
b=55.84       # storing float value
c=4+9j        # storing complex number
```

II. EXAMPLE OF NUMERICAL DATA TYPES

1. SOURCE CODE

```
# define the variables
a=45
b=77.83
c=5+9j
# print the types of variable using built-in method type()
r1=type(a)
r2=type(b)
r3=type(c)
# display the result
print("----------------------------------------")
print("\t Numerical Types - Example")
print("----------------------------------------")
print("a is : ",r1)
print("b is : ",r2)
print("c is : ",r3)
```

2. OUTPUT

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\IDE Projects\VSC Projects\Python Lab\ptmech> py t1.py
----------------------------------------
         Numerical Types - Example
----------------------------------------
a is :  <class 'int'>
b is :  <class 'float'>
c is :  <class 'complex'>
PS E:\IDE Projects\VSC Projects\Python Lab\ptmech>
```

3. Boolean Types

- This type of the data is used to store the Boolean values like True or False.

Built-in Data Type

bool

Example

```
flag=True      # storing Boolean value
flag=False     # storing Boolean value
```

4. None Type

- It is a special data type is used to store a null value / indicating the absence of a value.
- In python, the **None** keyword is an object and it is a data type of the built-in class **None**

Built-in Data Type

None

NOTE

- None is **not same** as boolean value **False**
- It is **not equal to 0**. It simply null.
- It is **not an empty string**
- Comparing anything to None type gives False always (except None itself)

Example

```
a=None          # here variable a is defined as null object
```

5. Binary Types

- It represents the **binary data**
- In python, the **bytes**, **bytearray** and **memoryview** are built-in classes / types which are used to work with binary data and memory views of binary data
- This type is used for handling tasks like handling binary files, network communication and low level data manipulation
- It is classified as three types. They are
  - bytes
  - bytearray
  - memoryview

bytes

- It is an immutable type which represents the sequence of bytes (8 bits)
- Each element in a bytes object is an integer object which is ranging from 0 to 255.
- It is mainly used for handling binary data like reading / writing files, network communication
- In python, the byte type is called as **bytes**.
- In python, the byte is defined by **b as the prefix** followed by sequence of bytes enclosed in single or double quotes.

Example

```
d1=b'welcome'

d2=b"good morning"
```

bytearray

- It is similar to bytes type
- **Unlike bytes type**, this type of the data can be **modified after creation**.
- This object is mainly used for modifying the binary data in place like processing binary files or network protocols
- This object type can be created by using the bytearray() constructor which accepts the list of values.

Syntax

```
variable-name=bytearray([v1, v2, … vn])
```

Where,

variable-name is an user defined name

bytearray is a reserved keyword

[ ] operator indicates array

v_1, v_2, … v_n are list of values

Example

```
lt=bytearray([12, 45, 99, 72, 81])
```

memoryview

- This type is used to create a view of memory containing binary data
- It does not store the data itself but provides a view into memory where the binary data is stored.
- This is best for handling large amounts of data without copying it.

Example

```
lt=bytearray([12, 45, 99, 72, 81])

mv=memoryview(lt)
```

III. BYTE EXAMPLE

1. SOURCE CODE

```
print("-------------------------------------")
print("\t Bytes Type Example")
print("-------------------------------------")
# define the bytes variables with initial values
g=b'Hi'
y=b"Good Morning"
# display the bytes values
print(g)
print(y)
# display the bytes type
print("Type 1: ",type(g))
print("Type 2: ",type(y))
```

2. OUTPUT

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\IDE Projects\VSC Projects\Python Lab\ptmech> py byte.py
--------------------------------------
          Bytes Type Example
--------------------------------------
b'Hi'
b'Good Morning'
Type 1:  <class 'bytes'>
Type 2:  <class 'bytes'>
PS E:\IDE Projects\VSC Projects\Python Lab\ptmech>
```

IV. BYTEARRAY AND MEMORYVIEW EXAMPLE

1. SOURCE CODE

```
print("-----------------------------------")
print("\t Memory View Type")
print("-----------------------------------")
# definition of byte array and memoryview objects
lb=bytearray([12,44,55,66,81])
# display the memory of byte array via memoryview object
mm=memoryview(lb)
print(mm)
```

2. OUTPUT

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\IDE Projects\VSC Projects\Python Lab\ptmech> py bt.py
------------------------------------
          Memory View Type
------------------------------------
<memory at 0x000002024D645180>
PS E:\IDE Projects\VSC Projects\Python Lab\ptmech>
```

6. Sequence Types

- This type of the data is used to **store the collection of items**
- It is classified as three types. They are
  - list
  - tuple
  - range

Built-in class type

list, tuple, range

Example 1 – List Type

```
ld=[12,25,42,72,81]
```

Example 2 – Tuple Type

```
td=(12,25,42,72,81)
```

Example 3 – Range Type

```
rd=range(1,9)          # represents the sequential of 1 to 8
                         numbers. (9 is Exclusive)
```

7. Mapping Type

- This type of the data is used to store the data in the form of key-value
- It is defined using the braces {}
- The mapping type in python is called as **dict**.

Built-in class type

```
dict
```

Example

```
# store details of student
stu={1:"rohit", 5:"kholi", 3:"ram",15:"raja"}
```

8. Set Types

- This type of the data is used to store the unordered collection of unique items
- It is defined using the braces {}
- It is classified as two types. They are
  - set
  - frozenset

Built-in class types

```
set, frozenset
```

Example 1 – using set type

```
# store details of phones using set type
st={"oppo a79", "nokia g42", vivo t2 pro", "honor 90"}
```

Example 2 – using frozenset type

```
# store details of phones using frozenset type
ft=frozenset{"oppo a79", "nokia g42", vivo t2 pro", "honor 90"}
```